

SOC/L1 - 3.3. Splunk: The Basics

Splunk: The Basics

1 - Introduction

Splunk is a platform for collecting, storing, and analysing machine data. It provides various tools for analysing data, including search, correlation, and visualisation. It is a powerful tool that organisations of all sizes can use to improve their IT operations and security posture.

Splunk is one of the leading SIEM solutions in the market. It allows users to collect, analyze, and correlate network and machine logs in real time. In this room, we will explore the basics of Splunk and its functionalities, and how it provides better visibility of network activities and helps speed up detection.

2 - Splunk Components

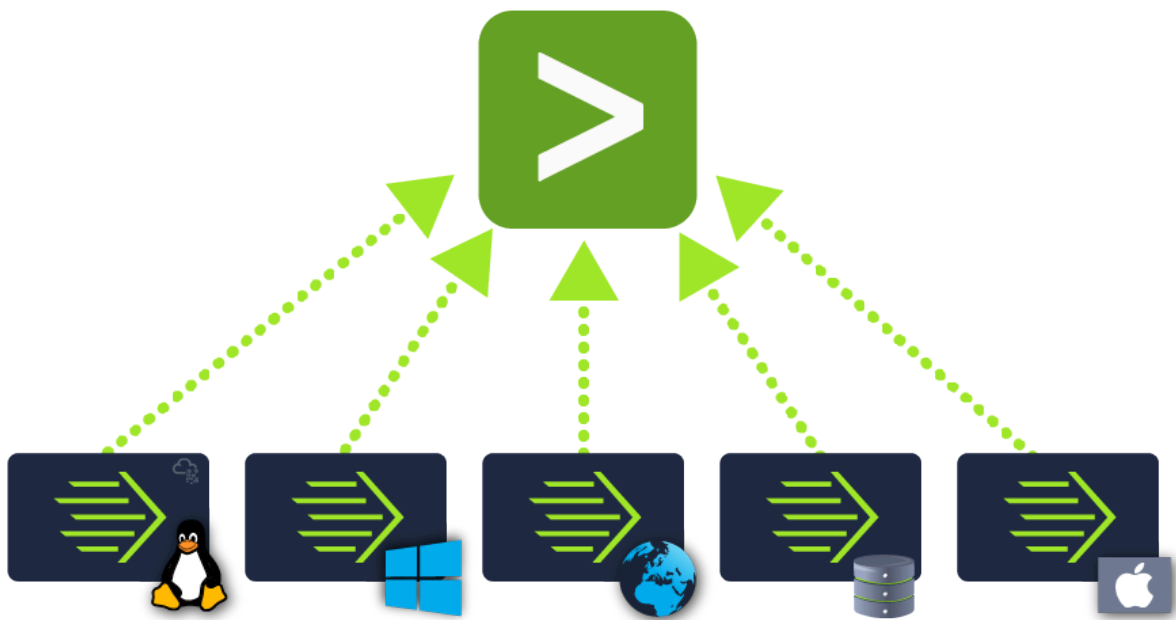
Splunk has three main components: Forwarder, Indexer, and Search Head. These components work together to help us search and analyze the data. These components are explained below:



1. Splunk Forwarder is a lightweight agent installed on the endpoint intended to be monitored, and its main task is to collect the data and send it to

the Splunk instance. It does not affect the endpoint's performance as it takes a few resources to process. Some of the key data sources are:

- Web server generating web traffic.
- Windows machine generating Windows Event Logs, PowerShell, and Sysmon data.
- Linux host generating host-centric logs.
- Database generating DB connection requests, responses, and errors.



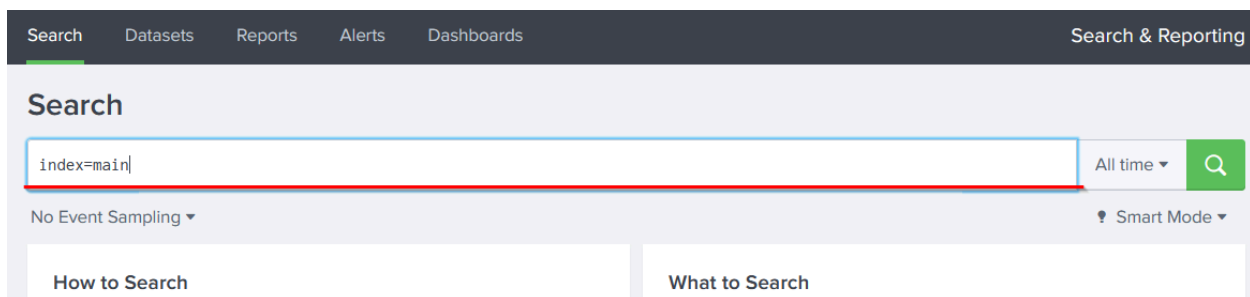
The forwarder collects the data from the log sources and sends it to the Splunk Indexer.

2. Splunk Indexer plays the main role in processing the data it receives from forwarders. It parses and normalizes the data into field-value pairs, categorizes it, and stores the results as events, making the processed data easy to search and analyze.

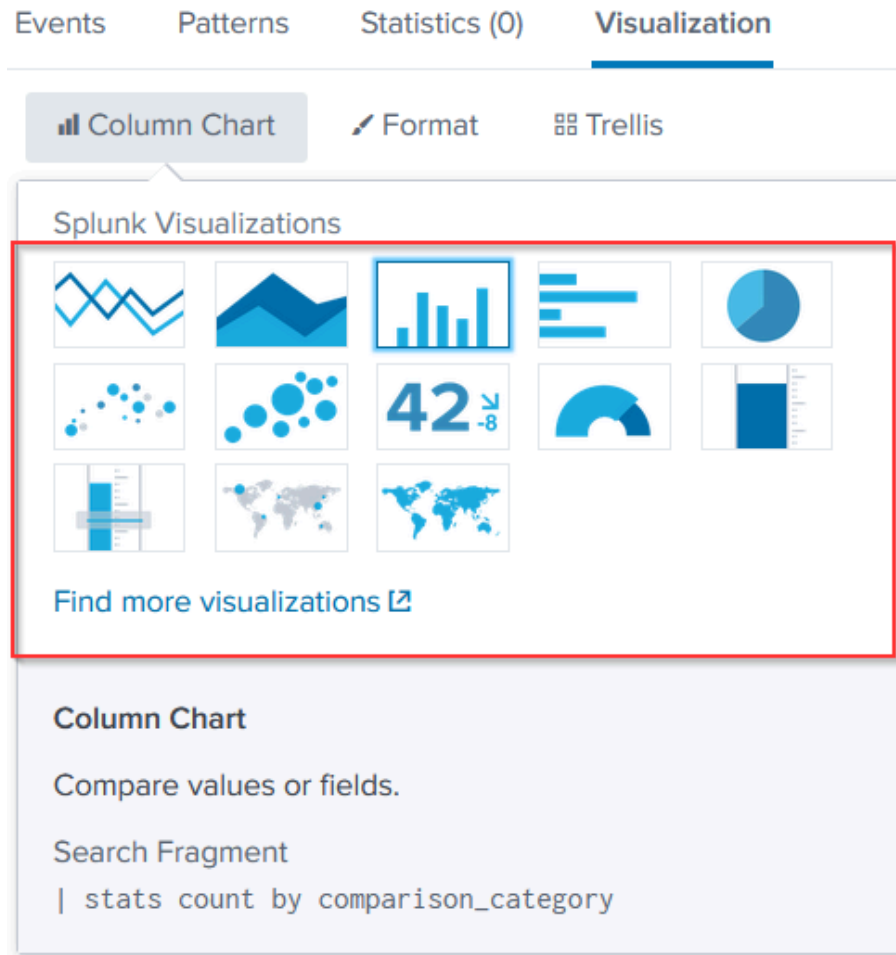


Now, the data, which is normalized and stored by the indexer, can be searched by the Search Head, as explained below.

3. Search Head: Splunk Search Head is the place within the **Search & Reporting App** where users can search the indexed logs, as shown below. The searches are done using the **SPL** (Search Processing Language), a powerful query language for searching indexed data. When the user performs a search, the request is sent to the indexer, and the relevant events are returned as field-value pairs.



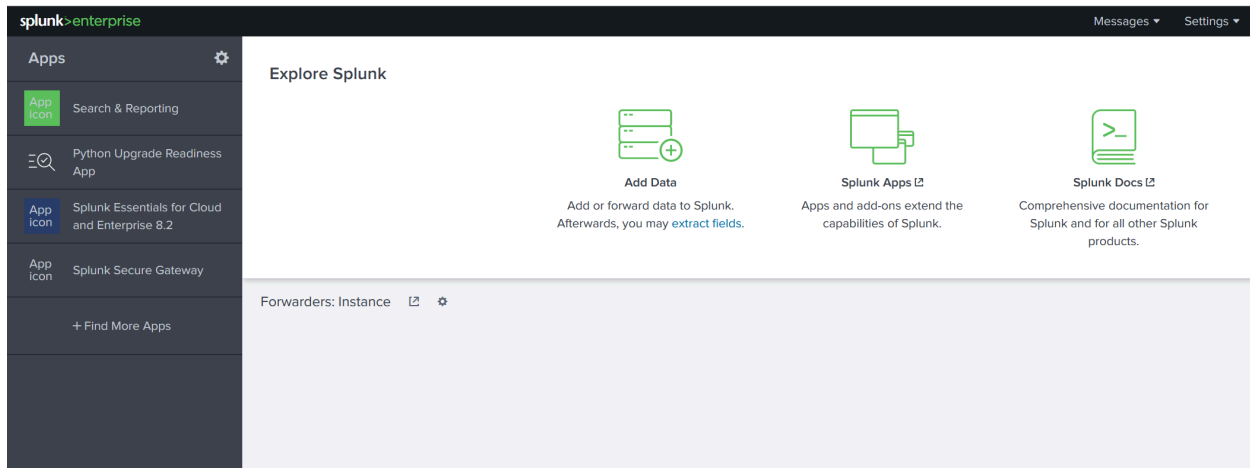
The Search Head also allows you to transform results into presentable tables and visualizations such as pie, bar, and column charts, as shown below:



Which component is used to collect and send data over the Splunk instance?
Forwarder

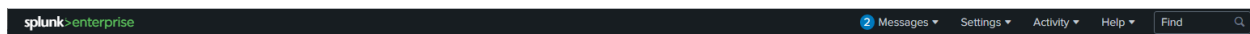
4 - Navigating Splunk

When you access Splunk, you will see the default **home screen** as shown below:



Let's look at each section of this home screen.

Splunk Bar: The top panel is the **Splunk Bar** as shown below:

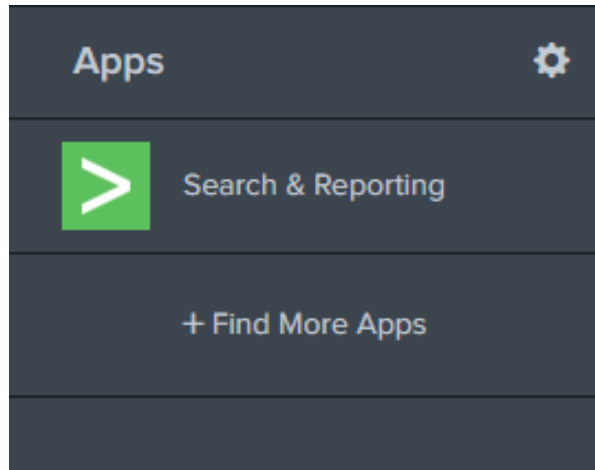


In the Splunk Bar, we have the following options available:

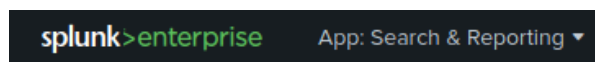
- **Messages:** View system-level notifications and messages.
- **Settings:** Configure Splunk instance settings.
- **Activity:** Review the progress of search jobs and processes.
- **Help:** View tutorials and documentation.
- **Find:** Search across the App.

The Splunk Bar, allows users to switch between installed Splunk apps instead of using the Apps panel.

Apps Panel: Next is the **Apps Panel**. This panel shows the apps installed for the Splunk instance. The default app for every Splunk installation is **Search & Reporting**.

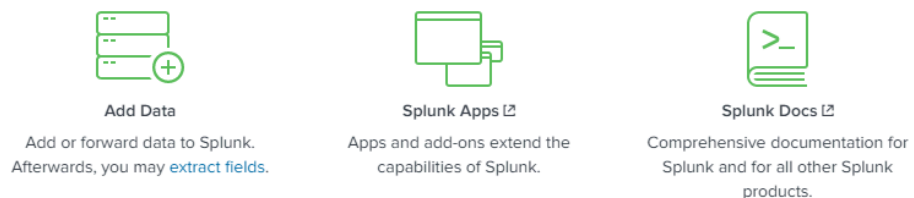


You can also switch between the Splunk Apps directly from the Splunk Bar, as shown below, without using the Apps Panel.

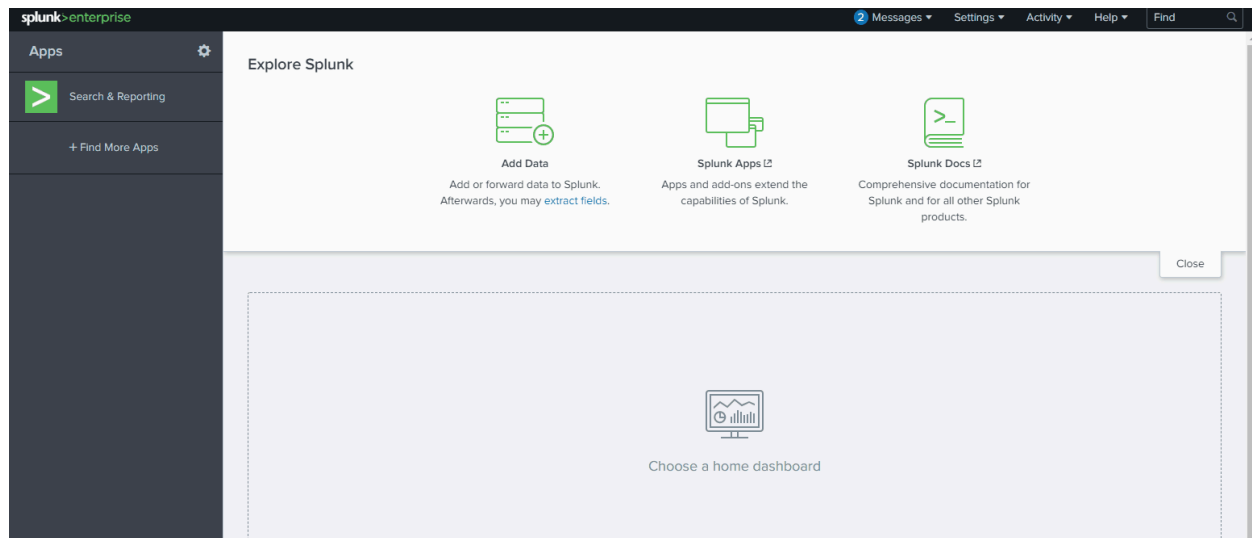


Explore Splunk: The next section is **Explore Splunk** . This panel contains quick links to add data to the Splunk instance, add new Splunk apps, and access the Splunk documentation.

Explore Splunk



Splunk Dashboard: The last section is the **Home Dashboard**. By default, no dashboards are displayed. You can choose from a range of dashboards readily available within your Splunk instance. You can select a dashboard from the dropdown menu or by visiting the **dashboards listing page**.



You can also create dashboards and add them to the Home Dashboard. The dashboards you create can be viewed separately from the other dashboards by clicking on the **Yours** tab.

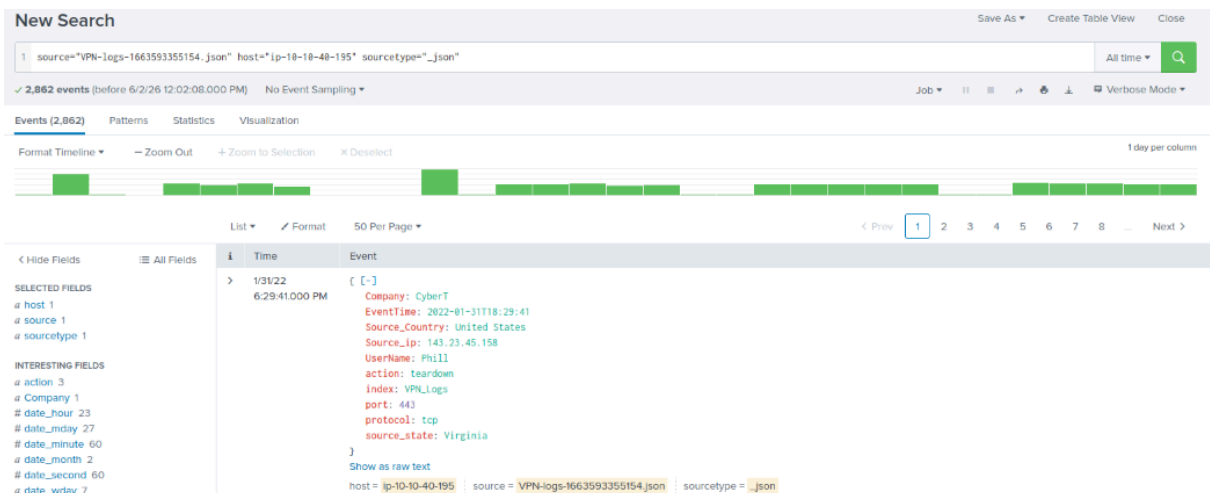
In the Add Data tab, which option is used to collect data from files and ports?
Monitor

5 - Adding Data

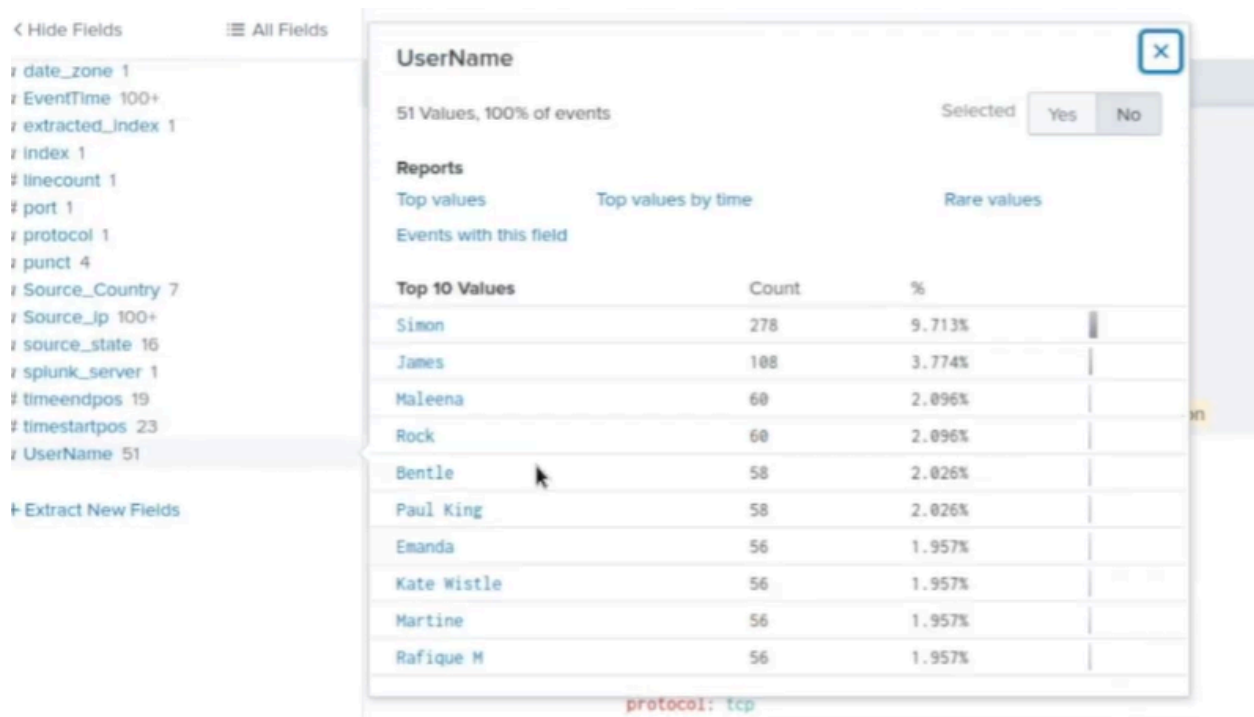
To upload the data successfully, you must follow five steps, which are explained below:

1. **Select Source:** Choose the Log file and the data source.
2. **Select Source Type:** Select what type of logs are being ingested, e.g, JSON, syslog.
3. **Input Settings:** Select the index where these logs will be dumped and the HOSTNAME to be associated with the logs.
4. **Review:** Review all the configurations.
5. **Done:** Complete the upload. Your data will be uploaded successfully and ready to be analyzed.

Upload the data attached to this task and create an index "VPN_Logs". How many events are present in the log file? 2862



How many log events are captured by the user Maleena? 60



What is the username associated with IP 107.14.182.38? Smith

The screenshot shows a Splunk search interface. On the left, a list of fields is visible, including 'date_year', 'date_zone', 'EventTime', 'extracted_index', 'index', 'linecount', 'port', 'protocol', 'punct', 'Source_Country', 'Source_ip', 'source_state', 'splunk_server', 'timeendpos', 'timestartpos', and 'UserName'. The 'UserName' field is selected. On the right, a 'Reports' panel is open, showing a table with the following data:

Values	Count	%
Smith	26	100%

searchsource="VPN-logs-1663593355154.json" Source_ip="107.14.182.38"

What is the number of events that originated from all countries except France?

2814

The screenshot shows a 'New Search' page in Splunk. The search query is: `source=*VPNlogs.json* host=*ip-10-10-40-195* sourcetype=*_json* NOT Source_Country=France`. The results show 2,814 events (before 9/12/25 1:21:57.000 PM). Below the search bar, there are tabs for 'Events (2,814)', 'Patterns', 'Statistics', and 'Visualization'. The 'Events (2,814)' tab is selected, and a timeline visualization is shown. Below the timeline, there are buttons for 'Format Timeline', 'Zoom Out', 'Zoom to Selection', and 'Deselect'. At the bottom, there is a table with columns for 'Time' and 'Event'.

search: source="VPN-logs-1663593355154.json" NOT Source_Country=France

How many VPN events were associated with the IP 107.3.206.58? 14

Search | Analytics | Datasets | Reports | Alerts | Dashboards

New Search

source="VPNlogs.json" host="ip-10-10-40-195" sourcetype="_json" Source_ip="107.3.206.58"

14 events (before 9/12/25 1:22:35.000 PM) No Event Sampling ▼

Events (14) | Patterns | Statistics | Visualization

Format Timeline ▼ | — Zoom Out | + Zoom to Selection | × Deselect

List ▼ | Format | 50 Per Page ▼

Hide Fields	All Fields		Time	Event
SELECTED FIELDS		>	1/31/22 5:50:18.000 PM	{ [-] Company: CyberT EventTime: 2022-01-31T17:50:18

source="VPN-logs-1663593355154.json" Source_ip="107.3.206.58"